

NW-200 Autonomous Warehouse Robot - Operator Manual

This manual describes the safe operation and routine maintenance of the NW-200 autonomous mobile robot, manufactured by Northwind Robotics. Read it fully before operating the robot. Keep it available to every operator on site.

The NW-200 moves shelving units and pallets inside warehouses. It navigates with a 360-degree lidar sensor and two depth cameras, and follows routes assigned by the Northwind Fleet Manager software.

Technical specifications:

Maximum payload: 150 kg.

Maximum speed: 1.8 metres per second (limited to 1.0 m/s in zones marked for pedestrians).

Battery: 48 V lithium iron phosphate, rated runtime of 8 hours per charge under typical load.

Charging time: 90 minutes from 10% to 100% on the NW-C2 docking charger.

Operating temperature: 0 to 40 degrees Celsius.

Weight without payload: 210 kg.

Safety

Only operators who have completed the Northwind operator training may start, stop or reassign an NW-200.

The red emergency stop buttons on each side of the robot cut power to the drive motors immediately. After an emergency stop, the robot can only be restarted from the Fleet Manager console, after an operator has confirmed that the area is clear.

Never ride on the robot, and never place a payload that blocks the front lidar sensor. Keep a minimum clearance of 0.5 metres around a moving robot.

The robot must not be operated on ramps steeper than 5 degrees or on wet floors.

Error codes

When the status light turns amber or red, the Fleet Manager console shows an error code. The most common codes are:

E-104 Lidar obstruction: the lidar field of view is blocked. Remove the obstruction and clean the lidar window with a dry microfibre cloth. The robot resumes automatically.

E-221 Battery overheating: battery temperature above 55 degrees Celsius. The robot stops and must cool down for at least 30 minutes. If E-221 occurs twice within one week, contact field service; do not keep operating the robot.

E-310 Wheel motor stall: a drive wheel is blocked, often by shrink wrap or straps. Press the emergency stop, remove the debris, then restart from the console.

E-502 Localisation lost: the robot cannot match its position to the site map. Push it manually to the nearest marked docking point and restart it there.

Maintenance schedule

Daily: inspect the lidar window and depth cameras for dust, and check the wheels for wrapped debris.

Every 500 operating hours: inspect the drive wheels for wear and replace them if the tread depth is below 3 mm. Check the tightness of the payload lift bolts.

Every 2,000 operating hours: replace the lift actuator seals and run the battery health diagnostic from the Fleet Manager.

Batteries should be replaced when their capacity falls below 70% of the original rating, which is typically after about 3,000 charge cycles.

Warranty: the NW-200 has a 24-month warranty covering parts and labour. The warranty is void if maintenance intervals are not followed or if non-Northwind parts are fitted.